AWS – S3- Glacier - CloudFront

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AWS – S3
Simple Storage service

Benefits

Durable

Integrated

Low Cost

Available

Secure

High Performance

Scalable

Easy to Use

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Use Cases

Content Storage & Distribution

Big Data Analytics

Static Website hosting

Cloud-native Application Data

Disaster recovery

Backup & Archiving

Accessed via API's

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Agenda

Amazon S3 Concepts & Fundamentals

Namespaces

Access Controls

Storage Classess

Encryption and Security Features

Versioning & Cross-Region replication

LifeCycle Rules

Website Hosting

Audit Logs

Time-Limited Access to Objects

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Concepts of S3

- Bucket
  - It is like a Containers
  - Bucket name has to be UNIQUE across all the regions of AWS and all accounts.
    - Eg:- Like your Gmail ID
  - By Default max of 100 Buckets can be create per Account.
  - Each Object has an limitation of 5TB.
- Objects
  - Entities Stored in S3
  - Consist of Data and Metadata
  - Metadata is a set of name-value pairs that describe the object
  - Object is uniquely identified within a bucket by a Key and a Version ID.
- Files and Folders can be uploaded and downloaded Directly from the Web links.
- There is no Dedicated instance created for storing the data.
- Buckets are created on Region Level.

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Storage Classes: Comparing Durability and Availability

Storage Class	Durability (designed for)	Availability (designed for)	Other Considerations
STANDARD	99.999999999%	99.99%	None
STANDARD_IA	99.999999999%	99.9%	There is a retrieval fee for STANDARD_IA objects. This class is most suitable for infrequently accessed data. For pricing information, see Amazon S3 Pricing.
ONEZONE_IA	99.999999999%	99.5%	Not resilient to the loss of the Availability Zone.
GLACIER	99.999999999%	99.99% (after you restore objects)	GLACIER objects are not available for real-time access. You must first restore archived objects before you can access them. For more information, see Restoring Archived Objects.
RRS	99.99%	99.99%	None

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Storage Class Pricing

https://aws.amazon.com/s3/pricing/

Storage Pricing (varies by region)

Region: US East (Ohio)

	Pricing
S3 Standard Storage	
First 50 TB / Month	$0.023 per GB
Next 450 TB / Month	$0.022 per GB
Over 500 TB / Month	$0.021 per GB
S3 Standard-Infrequent Access (S3 Standard-IA) Storage	
All storage	$0.0125 per GB
S3 One Zone-Infrequent Access (S3 One Zone-IA) Storage	
All storage	$0.01 per GB
Amazon Glacier Storage	
All storage	$0.004 per GB

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Options in S3

- Can Enable Versioning
- Encryptions – AWS or KMS
- Read/Write access to the Bucket
- Read/Write access to the Files
- Server Logging

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AWS – S3 – Web Hosting

2 Types of Hosting is available

- Client based Hosting
- Server based Hosting

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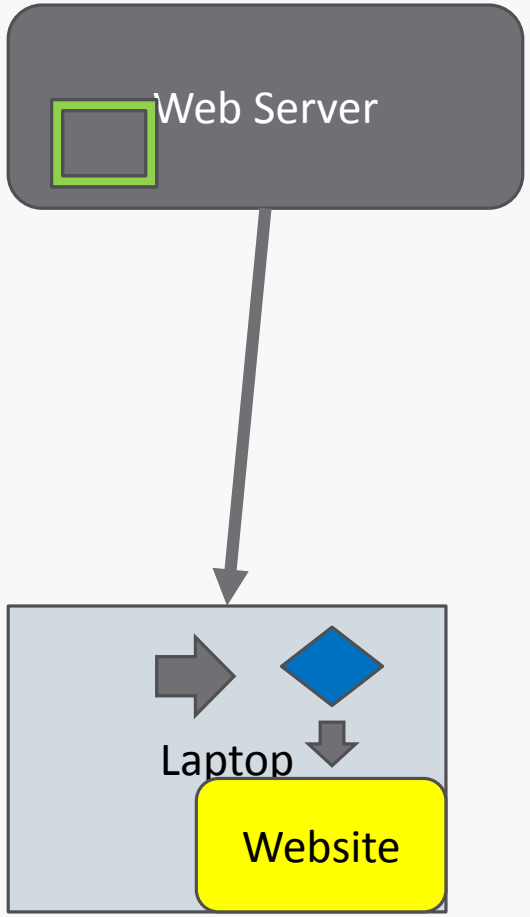
Client based Hosting



- The website page gets downloaded to the Laptop
- Then the page is proceed and shown on the screen.

- Typical example
  - Html pages
  - Videos
  - Audios

- Means, the data is only stored in the central webserver

- Processing of that data happens on the machine that is accessing it

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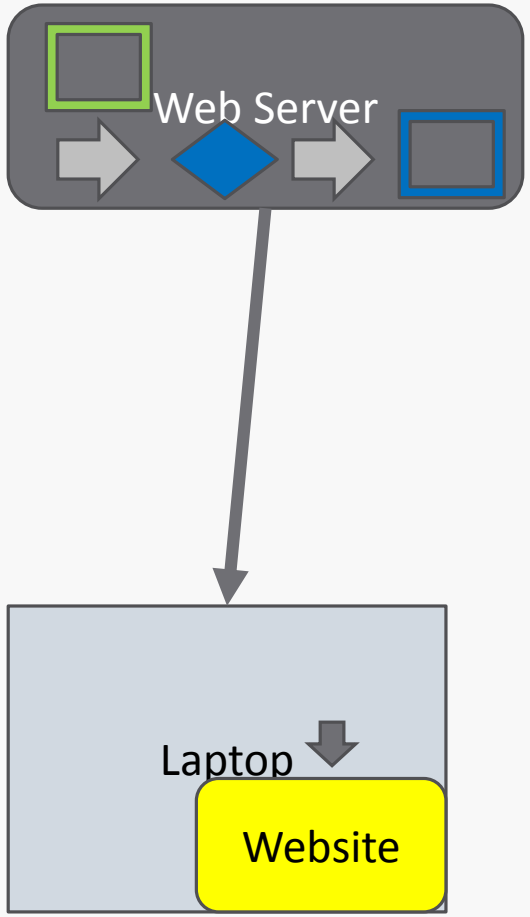
Server based Hosting



- The script file is processed on the server itself
- Only the output is sent to the machine that is accessing it.

- Typical example
  - Java based
  - .net
  - Python
  - php

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S3- Web Hosting

- You can host a static website on Amazon Simple Storage Service (Amazon S3)
- They might also contain client-side scripts such HTML, Node.js.

- Amazon S3 **does not** support server-side scripting such as such as PHP, JSP, or ASP.NET.

AWS – S3 – Replication

S3- Replication

- It is a copy of the existing S3 Bucket
- Replication of S3 bucket CANNOT be done in the same Region
- Either complete Bucket can be replicated or based on some “Prefix” values data can be replicated.
- Replication can be done to a Different Account S3 Bucket as well.
  - Create a Bucket Policy on the destination Bucket to allow the source aws account to copy the data.

- Replication is a backup of the Source Bucket.
  - If the file is deleted on the Source bucket , it would NOT be deleted on the destination bucket automatically.

- Replication is effective to the **files uploaded** after Enabling the Replication feature

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S3 – Replication - Pricing

Cross-Region Replication Pricing

You pay the Amazon S3 charges for

Storage

Requests

Inter-region data transfer for the replicated copy of data in addition to the storage charges for the primary copy.

Pricing for the replicated copy of storage is based on the destination region, while pricing for requests and inter-region data transfer are based on the source region.

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AWS – Glacier

S3 - Lifecycle

- 3 Types of LifeCycle of S3 Bucket
  - The Data in S3 Can be moved to “Standard –IA”
  - The Data in S3 Can be moved to “Once-Zone – IA”
  - The Data in S3 Can be Archived(moved and OFFline data) to “**Glacier**”

- To transition the data to “**Standard-IA**” and “**one-Zone-IA**”, the data has to be min of “**30**” days old.

- To Archive the data to **Glacier**, the data can be min of “1” day old

- We can apply the rules either on “Current Version” or “Previous Version’s” of the files.

- Once the data is set to expire, we **cannot RETRIVE** the data after it is deleted.

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Glacier - Pricing

Storage pricing

Region: US East (N. Virginia)

- $0.004 per GB / Month

Retrieval pricing

Region: US East (N. Virginia)

Retrieval Time	Data Retrievals
Expedited	$0.03 per GB
Standard	$0.01 per GB
Bulk	$0.0025 per GB

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Retrieval request pricing

Region: US East (N. Virginia)

Retrieval Time	Retrieval Requests
Expedited	$10.00 per 1,000 requests
Standard	$0.05 per 1,000 requests
Bulk	$0.025 per 1,000 requests

*Expedited On-Demand requests, like EC2 On-Demand instances, are available the vast majority of the time.

Provisioned expedited retrieval

Region: US East (N. Virginia)

- $100.00 per Provisioned Capacity Unit**

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S3 – LifeCycle – Glacier – Output

Amazon S3 > vishwa19820715

Newwinkey.pem Latest version

Overview | Properties | Permissions

Open | Download | Download as | Make public | Copy path

Owner
vishwaawsac

Last modified
Apr 24, 2018 8:37:32 AM GMT+0530

Etag
9db3169f41edc6d3e2facf08cf5a394c

Storage class
Glacier

Server-side encryption
None

Size
1692

Link
https://s3.amazonaws.com/vishwa19820715/Newwinkey.pem

Warning
This object is in Glacier. You cannot download it or make it public. You cannot edit any property or permission.

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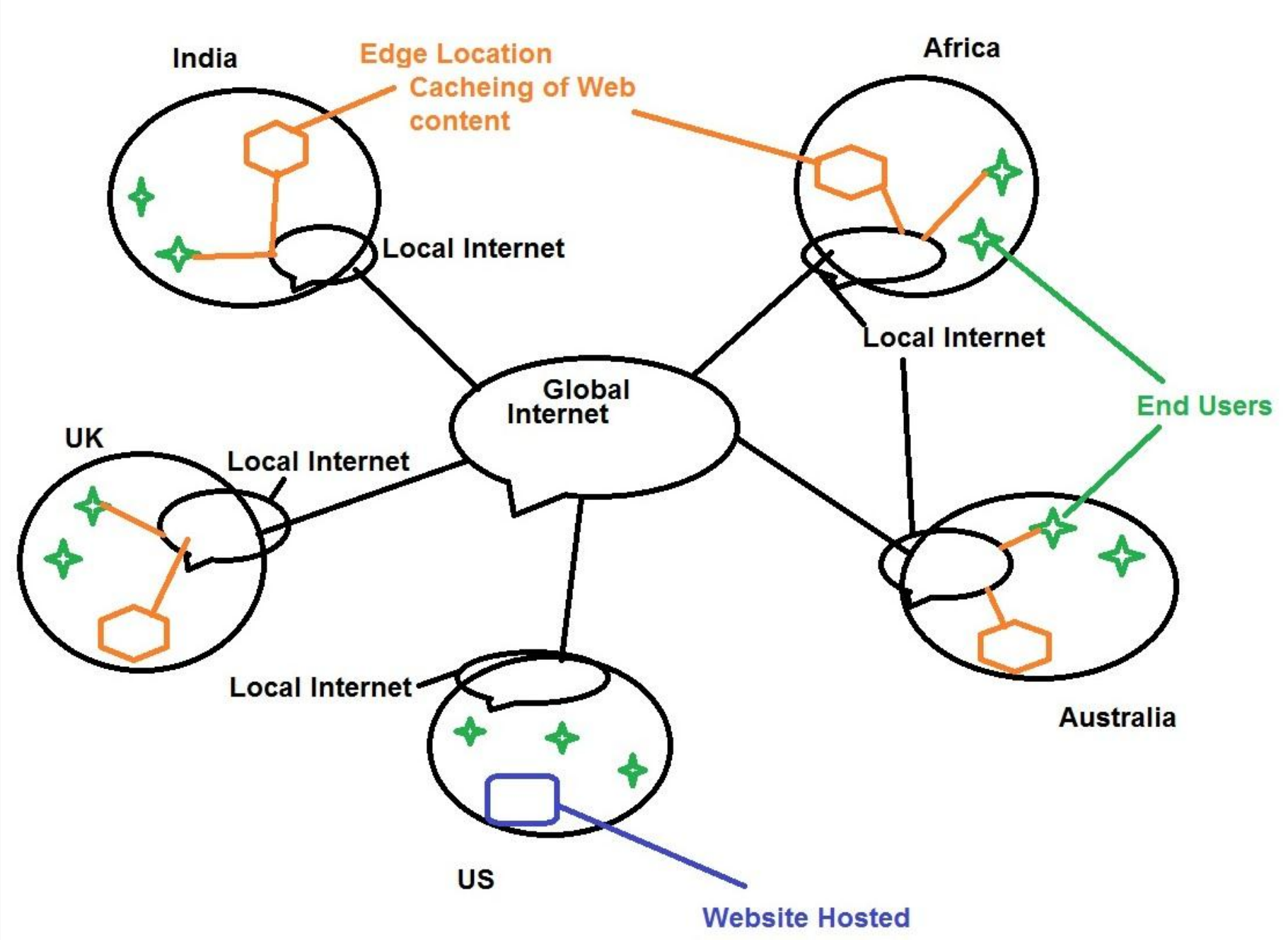
AWS – Cloud Front

Concepts of Content Delivery Network (CDN)



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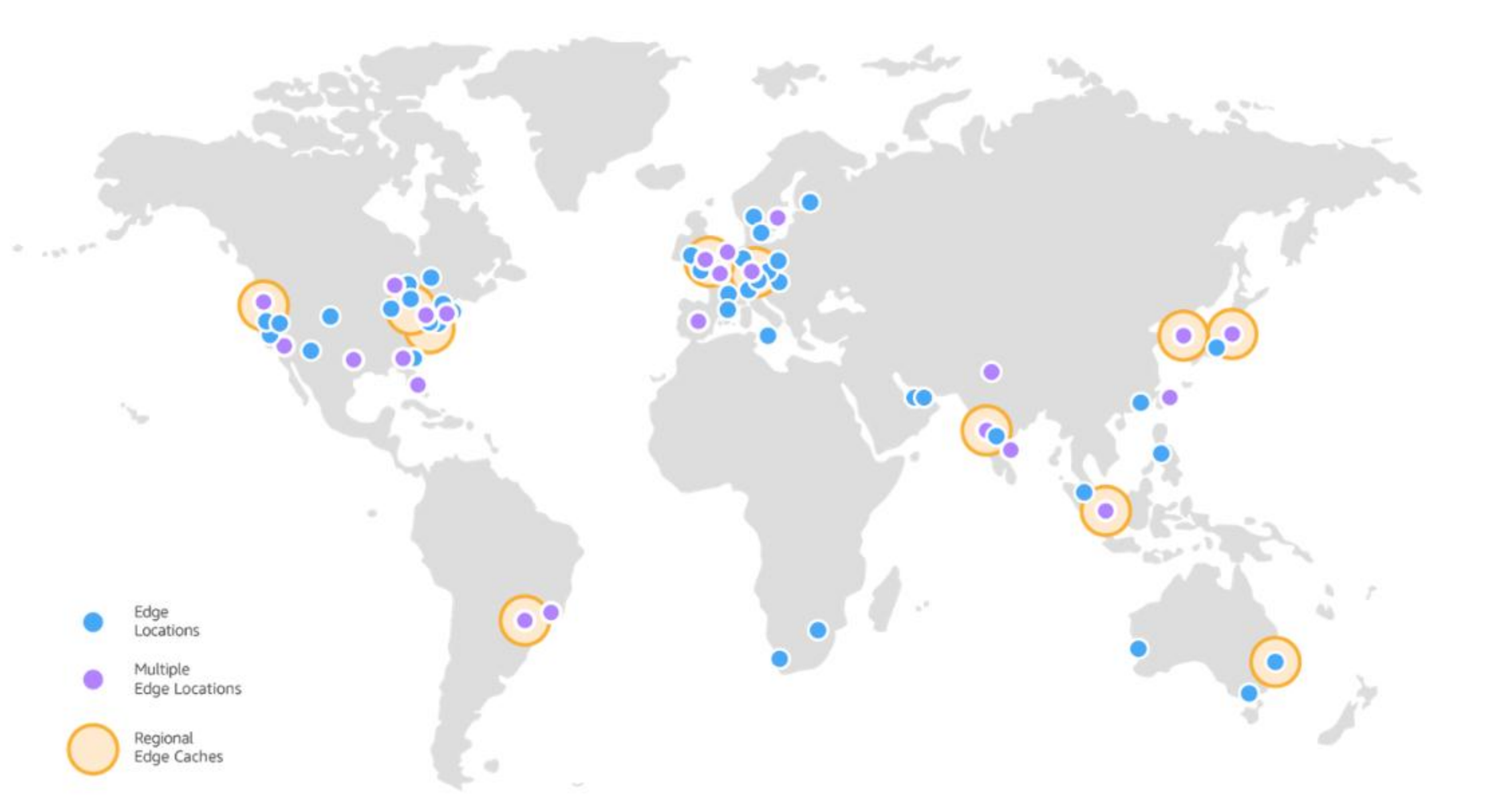
Concepts of CloudFront

- Amazon CloudFront is a fast content delivery network (CDN) service.
- Support's Data, Videos, Applications and API's.
- CloudFront is integrated with AWS – both physical locations that are directly connected to the AWS global infrastructure, as well as other AWS services.


Edge
Locations
Multiple
Edge Locations
Regional
Edge Caches

Amazon CloudFront Key Features

- Amazon CloudFront Infrastructure
- Security
- Protection against Network and Application Layer Attacks
- SSL/TLS Encryptions and HTTPS
- Access Control
- Compliance
- Availability
- Increase application availability
- Enabling redundancy for origins
- Performance
- Network optimizations for optimal performance
- Dynamic or static content
- Large libraries and media assets

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